Go in this order: Full reference document — Work through every question systematically and produce a single exhaustive .md file you can drop into your Notion/GitHub

Focused deep-dives — Pick 2-3 sections that are highest priority right now and go deep on those first (e.g., DSP architecture + building blocks, since that maps most directly to your Daisy Seed/TinyML work)
Interactive app — Build a Claude-powered synthesizer knowledge base artifact where you can query any section, with the content generated on demand
Layered approach — Start with a structured outline/taxonomy first so you can validate the scope, then fill in each section

I've created a single exhaustive Markdown reference document following the "full reference document + layered outline" approach, ready to drop into Notion or GitHub. It covers the entire taxonomy you listed (types, building blocks, architecture, DSP, UI/UX, genres, and use cases) in a structured, implementation-friendly way. [1] [2]

What's in the reference doc

The document starts with a taxonomy of synth types (analog, VA, FM, wavetable, sample-based, granular, physical modeling, modular, etc.), split into hardware vs software and polyphony/voice categories. [3] [1]
It then walks systematically through building blocks (oscillators, filters, envelopes, LFOs, modulators, sequencers, arpeggiators, effects), describing their canonical roles in both analog and digital instruments. [2] [1]

DSP & architecture sections

There are dedicated sections on high-level DSP signal flow, per-block DSP strategies (oscillators, filters, envelopes, LFOs, effects), and typical software architectures (plugin voice manager graphs, modular node graphs, and embedded/MCU layouts). [4] [3]
A final section sketches approaches to hardware emulation (component-level vs behavioural modeling, non-linearities, drift, and control-rate smoothing) that you can map directly onto Daisy Seed/TinyML or FPGA work. [5] [1]

How this matches your requested order

- **Full reference document:** Done — you now have a single .md reference that you can version-control and extend.
- **Layered approach:** The doc is structured as a taxonomy first, with each node containing concise notes and explicit pointers for future deep dives (e.g., sections 14–17 for DSP and architecture).

When you're ready, the next step can be:

- Pick 2–3 sections (I'd suggest the DSP architecture + building blocks cluster) and have me generate separate deep-dive docs with algorithms/pseudocode tuned to Daisy/embedded constraints.
- After that, we can design the "interactive app" layer — e.g., a small knowledge base schema plus prompt templates, so you can query any section from within your tooling.

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1. https://www.specdb.net/en/reviews/synthesizer-buying-guide-2026
2. https://rnbo.cycling74.com/explore/synth-building-blocks-intro
3. https://www.youtube.com/watch?v=9IJxoYbLf8o
4. https://psychosynth.com/sound-synthesis/types-of-synthesizers/software-101/
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7. https://muted.io/synth-list/
8. https://www.musicradar.com/news/best-synthesizers
9. https://www.slideshare.net/slideshow/week-6-assignment-48945920/48945920?nway-content_model=D
10. https://www.slideshare.net/slideshow/explain-the-usage-of-the-5-most-important-synthesis-modules-oscillator-filter-amplifier-envelope-and-lfo/18365014
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12. BC_Softsyn_Synthesizer.md
13. https://en.wikipedia.org/wiki/Software_synthesizer
14. https://www.youtube.com/watch?v=XtTlYBaueD0
15. https://www.youtube.com/watch?v=cqa3X-HwAnE
16. https://www.youtube.com/watch?v=XmI33zP51Us